LIGHTWEIGHT, COMPRESSION-ROBUST AUDIO DEEPFAKE DETECTION FOR EDGE DEPLOYMENT

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by

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DECLARATION

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CERTIFICATE

This is to certify that the work contained in this project report entitled "Lightweight, Compression-Robust Audio Deepfake Detection for Edge Deployment", submitted by **Nikkath Zabeen Shaik, Bhuvan S M, Abhijith Jayan, and Joshi Kshitij Santhosh** (Roll No: **2023BCS0060, 2023BCS0198, 2023BCS0109, and 2023BCS0087**) to the Indian Institute of Information Technology Kottayam in partial fulfilment of the requirements for the degree of **Bachelor of Technology in Computer Science and Engineering**, has been carried out by them under my supervision and has not been submitted elsewhere for the award of any degree.

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ABSTRACT

Generative audio systems can now synthesize speech, singing, music, and environmental sound with increasing realism, which threatens media integrity and trust in voice-based communication. A practical detector must work when the audio category is unknown at test time and when recordings have passed through lossy compression, resampling, noise, or narrowband channels that alter the spectral cues classifiers rely on. It must also be cheap to run: large self-supervised encoders achieve strong benchmark scores but are difficult to deploy on mobile or embedded hardware, which motivates compact models whose robustness is improved during training rather than by enlarging the inference graph.

This report investigates lightweight audio deepfake detection on the AT-ADD Track 2 benchmark. A MobileNetV3-Small baseline ($\sim$1.52 M parameters) operating on 80-bin log-mel features at 24 kHz is trained on clean audio only and compared with **Frequency-Time Robustness Coherence Learning (FT-RCL)**, a training framework that exposes the same backbone to paired clean and synthetically degraded waveforms. FT-RCL combines a dual classification loss, prediction coherence, eight-band feature signature coherence, a severity curriculum, and gated robustness weights, and it leaves the inference graph unchanged. A STAR-Edge extension, which adds asymmetric bona-fide robustness, severity trajectory coherence, and channel auxiliary detection, is also implemented. On 13,287 clean held-out clips, the baseline attains a Macro-F1 of 0.672 and an equal error rate (EER) of 0.216, compared with 0.659 and 0.234 for FT-RCL. FT-RCL improves Macro-F1 on environmental sound and music but yields no aggregate advantage on clean audio. A preliminary conditioned evaluation on 100 clips per model reports ΔMacro-F1 under resampling, noise, and telephony band-limiting; full-split ΔM analysis remains future work.

Keywords: audio deepfake detection; all-type audio; AT-ADD Track 2; robustness coherence learning; prediction coherence; log-mel spectrogram; MobileNetV3-Small; codec and channel degradation; severity curriculum; edge deployment; equal error rate.

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LIST OF ABBREVIATIONS

ABR	Asymmetric bona-fide robustness
ADD	Audio deepfake detection
ALLM	Audio large language model
AT-ADD	All-type audio deepfake detection (benchmark)
AUROC	Area under the receiver operating characteristic curve
CAD	Channel auxiliary detection
CE	Cross-entropy
CNN	Convolutional neural network
EER	Equal error rate
FNR, FPR	False-negative rate, false-positive rate
FT-RCL	Frequency-time robustness coherence learning
FTSC	Feature signature coherence
M0–M3	Training modes: clean-only baseline (M0), degraded-only (M1), dual-view with prediction coherence (M2), and the full FT-RCL objective (M3); see Table 3.3
PC	Prediction coherence
ROC	Receiver operating characteristic
SNR	Signal-to-noise ratio
SSL	Self-supervised learning
STC	Severity trajectory coherence
STFT	Short-time Fourier transform
TTM	Text-to-music
TTS	Text-to-speech

LIST OF SYMBOLS

$x,\ \tilde{x}$	Clean waveform and its degraded counterpart
y	Class label (0 = real, 1 = fake)
$f_s,\ N$	Sampling rate (24,000 Hz) and segment length in samples (96,000)
$g(\cdot\,; c)$	Waveform-domain degradation operator for condition c
$\phi(\cdot)$	Log-mel feature extractor
$\hat{\mathbf{L}}$	Normalized log-mel input to the network
$z,\ p$	Logits and class probabilities ($p = \mathrm{softmax}(z)$)
q	Fake-class probability, used as the detection score
$z_c,\ z_d,\ q_c,\ q_d$	Logits and fake-class probabilities of the clean and degraded views
$s_c,\ s_d$	Band-wise feature signatures of the clean and degraded views
K	Number of frequency subbands (8)
$\rho,\ \rho_{\max}(e)$	Degradation severity and the curriculum cap at epoch e
$\eta(\rho)$	Severity scaling factor of the M3 gate
$\alpha_t,\ \kappa$	Audio-type weight and prediction confidence of the M3 gate
$\bar{w}$	Batch-mean robustness weight
$\beta,\ \gamma_{\mathrm{pc}},\ \gamma_{\mathrm{ftsc}}$	Loss coefficients
$J,\ \lambda$	Robustness-aware selection score and its trade-off weight
ΔM	Metric drop, $M_{\mathrm{clean}} - M_{\mathrm{conditioned}}$

Chapter 1

Introduction

1.1 Background and motivation

Advances in neural text-to-speech (TTS), singing-voice synthesis, text-to-music systems, and general audio generative models have lowered the effort required to produce convincing synthetic audio. Such clips can be shared at scale through social media, messaging applications, and voice-based interfaces, where listeners often lack tools to verify provenance. **Audio deepfake detection (ADD)** is the learning problem of estimating whether a recording is *bona fide* (authentic) or *synthetic/spoof* (fake), usually from acoustic features alone.

Early community effort focused on attacks against automatic speaker verification, with shared datasets and standardized metrics such as the equal error rate (EER) [1], [2]. More recent benchmarks explicitly target **all-type** content (speech, singing, music, and environmental sound) and realistic channel perturbations [3]. This project adopts AT-ADD **Track 2**, in which the detector must classify real versus fake without knowing the audio category at test time. This setting is closer to open-world deployment than speech-only laboratory evaluation, but it also increases the risk that a model tuned on one domain fails on another.

Deployment adds a second motivation. Audio that reaches a detector has usually been re-encoded, resampled, or mixed with noise, and the detector is often expected to run on phones, browsers, or embedded devices with limited memory and energy. Self-supervised encoders with hundreds of millions of parameters dominate the top AT-ADD entries [3], but they are difficult to run on such hardware. These observations motivate a compact detector whose robustness to channel changes is obtained during training, so that the inference cost stays that of a small network.

1.2 Challenges in audio deepfake detection

Building a practical audio deepfake detector is difficult for several interacting reasons rather than because of binary classification alone. The main challenges addressed in this work are the following.

Diverse acoustic categories: Speech, singing, music, and environmental sound differ in harmonic structure, temporal dynamics, and signal-to-noise characteristics. Ar-

tifacts that indicate synthesis in speech (for example, phase inconsistencies in vocoded speech) may not transfer to music generated by diffusion or transformer-based models. A detector that exploits speech-specific cues can perform well on one audio type yet fail on music or environmental sound, so detection performance must be examined for each audio type rather than only in aggregate.

Generator diversity: Fake clips may originate from autoregressive TTS, flow-based vocoders, singing conversion, text-to-music, or general audio synthesis. Protocols that hold out generators at test time measure generalization to unseen synthesis pipelines [3]; a model may otherwise memorize generator-specific fingerprints rather than learn invariant signs of synthesis.

Robustness under realistic audio conditions: Benchmark clips are often stored in relatively high-quality formats, whereas deployed audio is frequently re-encoded, resampled for streaming, mixed with background noise, or filtered by telephone channels. Features that are useful on clean spectrograms may be attenuated or aliased after such processing, producing a **domain shift** between training assumptions and application conditions. Claims of compression robustness therefore require evaluation under explicit degradations, not clean scores alone.

Spurious cues: Classifiers can reach high accuracy by exploiting recording conditions, codecs, or dataset biases instead of genuine evidence of synthesis. Such shortcuts tend to break when the channel changes, so robustness must be evaluated explicitly rather than inferred from clean validation scores.

Computational constraints: Mobile, embedded, and browser-side inference limit parameter count, memory bandwidth, and energy per clip. Compactness is a necessary design constraint for edge use, but it is not sufficient: real-time suitability must be confirmed by measurement on a target platform.

1.3 Problem statement

Given a mono audio segment of fixed duration (four seconds at 24 kHz), estimate the label $y \in \{0, 1\}$ (real = 0, fake = 1). AT-ADD Track 2 mixes four audio types in evaluation, and the type is not provided at inference. The problem is to design a detector that (i) works across all four audio types, (ii) remains stable when the input has been degraded by compression, noise, resampling, or band-limiting, and (iii) keeps an inference cost compatible with edge deployment.

The **primary** experimental objective is clean held-out evaluation on AT-ADD Track 2 under a fixed, reproducible preprocessing and training protocol.

1.4 Objectives

The objectives of this project are to:

1. implement and validate a reproducible acoustic preprocessing pipeline and lightweight baseline architectures (pure 2D CNN vs. recurrent CRNNs) on speech (ASVspoof 2019) and multi-domain audio (AT-ADD Track 2), benchmarking against published literature;
2. design and implement Frequency-Time Robustness Coherence Learning (FT-RCL), which uses paired clean and degraded views, prediction coherence, band-wise feature signature coherence, a severity curriculum, and gated robustness weights, without changing the inference graph;
3. implement the STAR-Edge extension of FT-RCL, which adds asymmetric bona-fide robustness, severity trajectory coherence, and channel auxiliary detection;
4. evaluate all systems with Macro-F1, accuracy, AUROC, and EER on clean held-out audio, overall and per audio type, and log full training histories;
5. document checkpoint size and the planned edge-deployment measurements (latency, memory, INT8 accuracy).

1.5 Contributions

Methodological contributions:

- A unified coherence-based robustness training framework that jointly applies dual-view classification, prediction coherence, band-wise signature coherence, severity scheduling, and batch-mean robustness gating to a shared lightweight backbone, leaving the inference-time architecture identical to that of the baseline.
- A separation of waveform-domain degradation from the mel frontend parameters, which allows a controlled comparison between clean-only training and coherence-based training.
- A robustness-aware model selection criterion that balances clean development performance against the estimated metric drop under degradation.
- The STAR-Edge extension, which augments FT-RCL with asymmetric bona-fide robustness, severity trajectory coherence, and a channel auxiliary detection branch.

Experimental contributions:

- A foundational baseline investigation evaluating pure 2D CNN versus recurrent CRNN architectures on full-scale multi-domain audio, identifying duration shortcut artifacts on ASVspoof 2019, and establishing competitive performance against published literature (Pham et al., 2024).
- A clean-evaluation comparison of the baseline and FT-RCL on AT-ADD Track 2, documenting aggregate and per-type trade-offs, including the cases where FT-RCL does not outperform the baseline.
- Documented training and clean-evaluation protocols, with ablation and edge-benchmarking directions for follow-on work.

Chapter 2

Literature Review

This chapter synthesizes the curated reference set of the project. It proceeds from speech-centric spoofing benchmarks to all-type audio deepfake detection, then to robustness under channel distortion, alternative detector designs, and multimodal work that bounds the scope of this study. It closes with the research gap that motivates the proposed method.

2.1 Evolution of audio spoofing and deepfake detection

Research on fake audio began in speaker-verification security, where replay, synthesis, and voice-conversion attacks motivated shared corpora and standardized metrics such as the equal error rate (EER) [1], [2]. ASVspoof established rigorous logical-access evaluation for speech [2], but its early protocols were predominantly **speech-centric**. Surveys and systematic reviews document the widening threat surface as synthetic content spans speech, music, singing, and environmental sound [4]–[8]. Engineering-oriented reviews note that mel-spectrogram and CNN pipelines remain common baselines for investigators and practitioners [9], [10], while larger self-supervised encoders lead accuracy rankings at higher compute cost [11].

Generative modelling has increased the realism of TTS, vocoders, and general audio synthesis. Dataset-centric work such as WaveFake and FakeAVCeleb supports detection research on synthetic speech and multimodal forgeries [12], [13]. **FakeSound** targets *general* (not speech-only) fake audio detection and localization [14], motivating benchmarks in which music and environmental sound are first-class categories.

2.2 Benchmarks: from speech-only to all-type

AT-ADD (All-Type Audio Deepfake Detection) [3] addresses robust detection under realistic perturbations and type-agnostic evaluation across speech, singing, music, and environmental sound. Its Track 2 Macro-F1 is computed in two stages: F1 is averaged between real and fake *within* each audio type, and the result is then averaged across types, so strong speech-only performance is insufficient. Challenge systems report a large gap between compact baselines and winners that use heavy self-supervised

fusion [3], [15].

Recent all-type methods improve accuracy through audio large language models (ALLMs) and frequency-time reasoning: interpretable ALLM-based detection with reinforcement learning [16], dual-domain SSL fusion of EAT and XLS-R [15], and ALLM4ADD-style multimodal LLM prompting [17]. These approaches define the **research gap** for edge-scale systems: the present project keeps inference at MobileNetV3-Small scale [18] and studies train-time robustness instead of deploying 300M-parameter encoders at test time.

2.3 Feature representations and frontends

Compact ADD pipelines typically use log-mel or MFCC-like frontends with CNN classifiers [4], [10]. Similarly, Pham et al. evaluate linear filterbanks and dynamic LFCC features fine-tuned on off-the-shelf vision backbones including MobileNet-V3 [19]. Mcuba et al. compare MFCC, mel-spectrogram, chromagram, and raw spectrogram inputs with custom CNNs and VGG-16 and report architecture–frontend interactions that are relevant to the mel-CNN design choices in this project [10]. SSL-based detectors instead fine-tune Wav2Vec2, XLS-R, and related encoders; the Vicomtech ADD challenge system exemplifies robustness-oriented SSL fine-tuning under compression-like conditions [20]. Zhang et al. combine XLS-R with an SLS classifier [21], illustrating the accuracy–cost trade-off relative to fixed mel frontends.

2.4 Robustness to compression, noise, and channel change

Multiple surveys highlight the sensitivity of detectors to codecs, bandwidth limiting, reverberation, and background noise [5]–[7]. The Perturbed Public Voices (P2V) dataset explicitly supports robust ADD under public-voice perturbations [22], and cross-domain ADD benchmarks analyse generalization when training and test domains differ [23]. Imaging and multimodal studies discuss compression-aware forensic pipelines [24], which is consistent with the degradation families used in FT-RCL training (resampling, MP3 coding, additive noise, and telephony band-limiting).

Systematic reviews note that high in-dataset accuracy often fails to transfer under noise and adversarial manipulation [6], [25]. Rabhi et al. study adversarial attacks and countermeasures on audio-deepfake detectors [25], which supports the conservative robustness claims made in this report.

2.5 Consistency, trajectories, and alternative detectors

Beyond the standard classification loss, recent work explores embedding trajectories and partial-spoof localization. TRACE analyses speech foundation-model embedding trajectories for training-free partial deepfake detection [26]; it is conceptually related to the severity-aware and coherence-based training of FT-RCL, although it is not used at inference here. VoiceRadar proposes micro-frequency compositional cues for voice deepfake detection [27], an alternative inductive bias to mel-CNN features. Privacy-preserving ADD (SafeEar) decouples content understanding from spoof scoring [28], which corresponds to a different threat model from the binary AT-ADD Track 2 task.

2.6 Multimodal deepfake detection

Audio-visual fusion methods (AVoiD-DF, AVFF, X-AVDT, and joint audio-visual detection) achieve strong robustness by exploiting lip–speech consistency [29]–[32]. Hybrid and social-media-oriented frameworks further combine emotion or platform context [33]–[35]. **This project is audio-only** at inference, to match edge constraints and AT-ADD Track 2; the multimodal literature bounds the scope and motivates future cross-modal extensions.

2.7 Research gap

Synthesizing the above references, the following gaps motivate this project:

1. **Domain imbalance.** Lightweight mel-CNN detectors achieve relatively strong performance on speech and singing but often struggle on music and environmental sound in clean evaluation [3], [6], [14].
2. **Clean accuracy versus channel robustness.** Surveys and empirical studies report generalization and robustness failures under noise and compression [5], [7], [22], yet compact systems rarely report clean metrics and ΔM under one reproducible recipe.
3. **Inference budget mismatch.** All-type leaders rely on SSL fusion and LLM-based reasoning [15]–[17], which are poor fits for strict edge latency without compression or distillation that are not evaluated here.
4. **Integrated Coherence Training.** Optimizes clean/degraded classification, prediction and band-wise coherence, severity curriculum, and checkpoint selection using a fixed MobileNetV3-Small backbone (FT-RCL M3).

Chapter 3

Proposed Methodology

This chapter describes the end-to-end system. Section 3.1 gives the overall design, Section 3.2 formalizes the task, and Sections 3.3–3.5 cover the data, the log-mel frontend, and the degradation simulator. Sections 3.6–3.8 present the baseline, the proposed FT-RCL detector, and the STAR-Edge extension. Sections 3.9–3.11 describe optimization, the difference between training and inference, and the evaluation metrics.

3.1 System overview

The proposed system keeps a fixed, compact backbone (MobileNetV3-Small [18]) and improves its behaviour under channel changes only through the training objective. Figure 3.1 summarizes the workflow: waveforms are segmented, converted to normalized log-mel features, and classified; during robustness training each waveform is additionally degraded and passed through the same network, and the two views are tied together by coherence losses.

3.1.1 Design principles

- **Unchanged inference graph.** All robustness mechanisms act at training time; at test time the model computes one mel spectrogram and runs one forward pass.
- **Clean baseline for attribution.** A clean-trained baseline (mode M0, class-weighted cross-entropy on clean log-mel inputs only) uses identical preprocessing, data, and architecture, so that any effect of the auxiliary losses can be attributed to them.
- **Consistency learning.** Two views of the same instance, a clean waveform and a degraded variant with the same label, should map to similar outputs. Penalizing disagreement between the views counteracts unstable predictions and unstable internal features, whereas augmentation alone only exposes the model to degraded data without requiring agreement.

3.2 Problem formulation

Let $x \in \mathbb{R}^N$ be a mono waveform with sampling rate $f_s = 24{,}000\,\text{Hz}$ and fixed length $N = 4f_s = 96{,}000$ samples (four seconds). The label $y \in \{0, 1\}$ indicates real (0) or

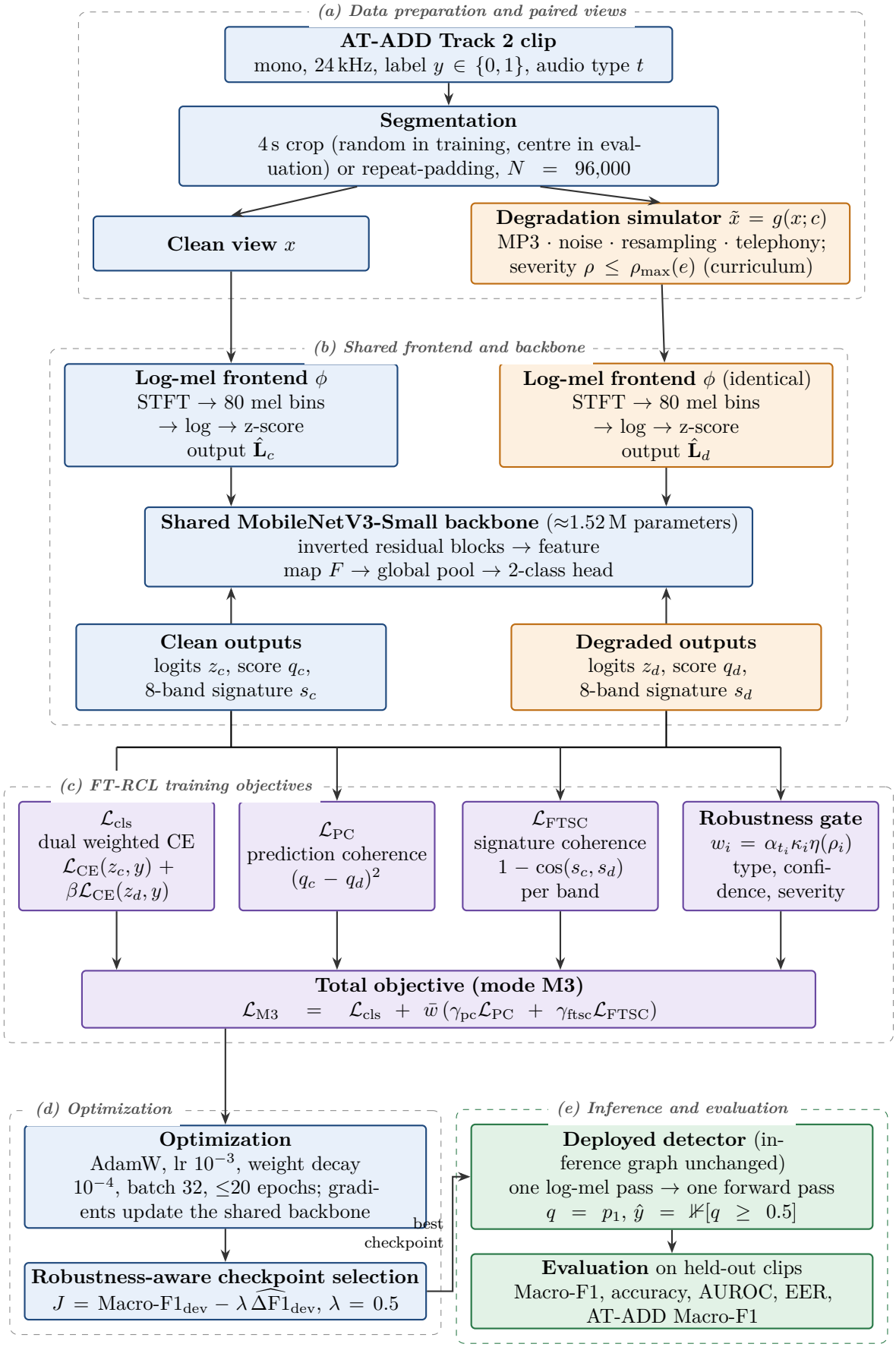


Figure 3.1. Overall end-to-end workflow of the proposed methodology. Orange blocks are used only during training; the deployed detector (green) keeps the single-view baseline inference graph.

fake (1). The classifier produces logits $z \in \mathbb{R}^2$ and probabilities $p = \mathrm{softmax}(z)$. The **detection score** used for ROC analysis and EER is the fake-class probability $q = p_1$.

For robustness training, a condition c from a finite catalogue defines a waveform-domain operator $g(\cdot\,; c)$. The paired sample $(x, \tilde{x})$ satisfies $\tilde{x} = g(x; c)$ with the same label y and the same temporal crop. Mel extraction $\phi(\cdot)$ is applied **identically** to both waveforms:

$$\mathbf{X} = \phi(x), \qquad \tilde{\mathbf{X}} = \phi(\tilde{x}). \tag{3.1}$$

Degradation never alters the mel hyperparameters or normalization statistics, which ensures a fair comparison with the clean baseline.

3.3 Dataset

3.3.1 AT-ADD Track 2

Experiments use **AT-ADD Track 2**, which provides binary real/fake labels with metadata for the audio type (speech, singing, music, sound) and, for fake clips, the synthesis generator. The benchmark defines separate train, development, and evaluation partitions, and the fake generators in evaluation are disjoint from those in training [3]. Real and fake clips are imbalanced in training (roughly eight real clips for every fifteen fake clips), which is handled by class weighting (Section 3.6).

3.3.2 Experimental data

Table 3.1 summarizes the size of the AT-ADD Track 2 benchmark, the held-out evaluation clips used for reported metrics, and the main training settings. The manifests contain 126,825 clips in total. Type counts in the evaluation subset are not balanced, so pooled metrics are dominated by speech.

3.4 Audio preprocessing and log-mel frontend

3.4.1 Loading and segmentation

Clips are resampled to f_s if necessary. If the duration exceeds four seconds, a crop of length N is taken (uniform random start in training, centre crop in evaluation). Shorter clips are extended by **repeat-padding** (tiling the waveform) rather than zero-padding.

Table 3.1. Dataset size, evaluation subset, and training settings.

Group	Quantity	Value
Full benchmark (clips)	Training split (16,000 real, 30,000 fake)	46,000
	Development split	36,536
	Evaluation split	44,289
	Total	**126,825**
Eval subset by audio type (clips)	Speech	8,083
	Singing	2,023
	Sound	1,794
	Music	1,387
	Total evaluated	**13,287**
Training settings	Batch size	32
	Random seed	42

3.4.2 Short-time Fourier transform and mel filtering

With a Hann window $w[n]$, frame length $L_w = 600$ samples (25 ms), hop $H = 240$ samples (10 ms), and FFT size $N_{\mathrm{FFT}} = 1024$ (the window is zero-padded to the FFT size),

$$X(m,k) = \sum_{n=0}^{L_w-1} x[n+mH]\, w[n]\, e^{-j2\pi kn/N_{\mathrm{FFT}}}, \qquad P(m,k) = |X(m,k)|^2. \tag{3.2}$$

Frames are centred using reflection padding, so a four-second segment yields $T_m = 1 + \lfloor N/H \rfloor = 401$ frames. A mel filter bank $\mathbf{W}_{\mathrm{mel}} \in \mathbb{R}^{80\times 513}$ with 80 bands (Slaney normalization) gives the mel power $\mathbf{P}_{\mathrm{mel}} = \mathbf{W}_{\mathrm{mel}}\mathbf{P}^{\top}$, and the log-mel features are

$$\mathbf{L} = \log(\mathbf{P}_{\mathrm{mel}} + \varepsilon), \qquad \varepsilon = 10^{-10}. \tag{3.3}$$

3.4.3 Global normalization

The mean μ and standard deviation σ are computed on the training split ($\mu \approx -9.19$, $\sigma \approx 6.85$ in the reference baseline) and applied in training and testing:

$$\hat{\mathbf{L}} = (\mathbf{L} - \mu)/\sigma. \tag{3.4}$$

The tensor $\hat{\mathbf{L}}$ is treated as a single-channel image of shape $1 \times 80 \times T_m$ by the CNN. Figure 3.2 illustrates the frontend.



Figure 3.2. Audio preprocessing and log-mel feature extraction.

3.5 Synthetic degradation (paired views)

3.5.1 Degradation catalogue

Degradations are applied in the waveform domain before ϕ. Table 3.2 lists the training catalogue; each condition has a severity level ρ that increases with the strength of the degradation. Figure 3.3 shows a clean and a degraded version of one clip in the waveform and log-mel domains. MP3 encoding requires an external codec tool on the host system. Each degradation records the condition family, severity level, and random seed, which are used by the curriculum and the gate.

Table 3.2. Degradation families and severity levels (training catalogue).

Family	Setting	Severity ρ
Resample	24 kHz → 16 kHz → 24 kHz	1
MP3 codec	128 / 64 / 32 kbps	2 / 3 / 4
Additive noise	SNR 20 / 10 / 5 dB	2 / 3 / 4
Telephony	Band-pass 300–3400 Hz	3

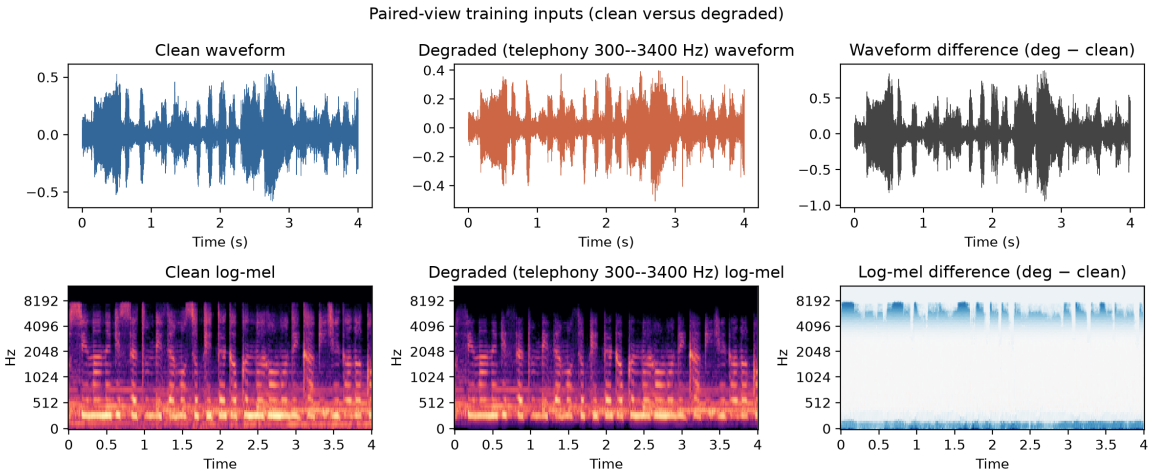


Figure 3.3. Clean versus degraded waveform and log-mel on the evaluation split (MP3 degradation is shown only when the codec tool is available).

3.5.2 Severity curriculum

Let $\rho_{\max}(e)$ be the maximum severity allowed at epoch e, with milestones at epochs 6 and 13:

$$\rho_{\max}(e) = \begin{cases} 1 & e < 6, \\ 3 & 6 \le e < 13, \\ 4 & e \ge 13. \end{cases} \tag{3.5}$$

Only conditions with $\rho \le \rho_{\max}(e)$ are sampled, so early training sees only the mildest degradation and the harder conditions of Table 3.2 enter gradually (Figure 3.4).

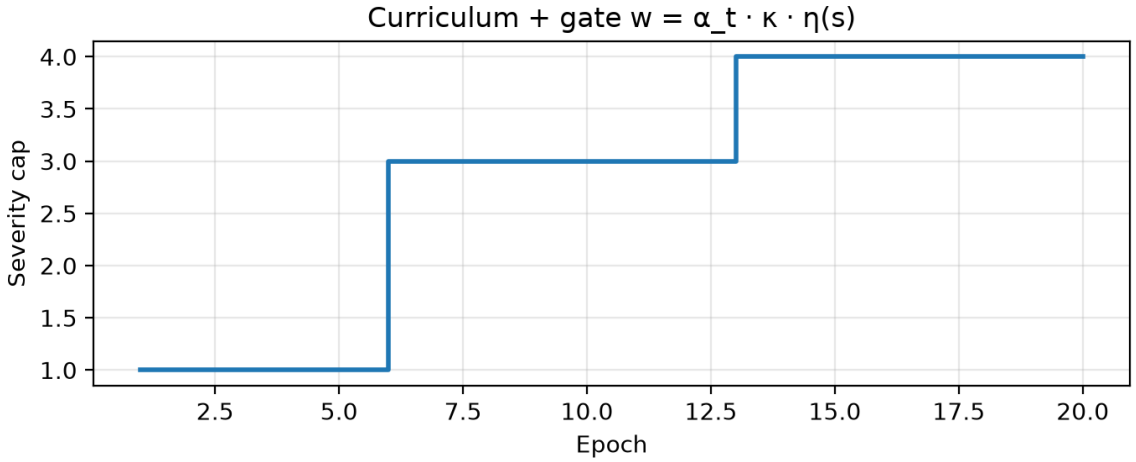


Figure 3.4. Severity curriculum of mode M3: maximum allowed severity $\rho_{\max}$ versus epoch.

3.5.3 Severity scaling

For a sample with severity ρ, the severity factor used by the M3 gate (Section 3.7.2) is

$$\eta(\rho) = \min\left(1,\ \frac{\rho}{\rho_{\max}(e)}\right). \tag{3.6}$$

3.6 Baseline model (mode M0)

3.6.1 Architecture

The baseline uses **MobileNetV3-Small** adapted to single-channel mel input (without ImageNet pre-training) and a two-logit classifier head, giving 1,519,618 parameters. The forward pass is $\hat{\mathbf{L}} \mapsto z = f_\theta(\hat{\mathbf{L}})$ (Figure 3.5). Its depthwise-separable convolutions and inverted bottlenecks keep the inference cost low. As demonstrated empirically in Section 4.1.2, pure 2D convolutional modeling with spatial global average pooling

outperforms recurrent CRNN architectures (LSTM/GRU) on multi-domain audio by over 2.2% EER, establishing MobileNetV3-Small as the optimal lightweight backbone for all-type audio.

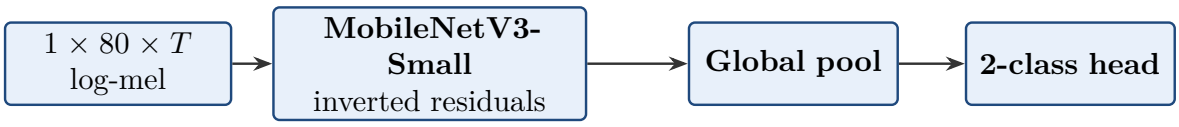


Figure 3.5. MobileNetV3-Small baseline on single-channel log-mel input (∼1.52 M parameters).

3.6.2 Class-weighted cross-entropy loss

With class counts n_0, n_1 and total $n = n_0 + n_1$, the weights $\omega_j = n/(2n_j)$ balance the real and fake classes:

$$\mathcal{L}_{\mathrm{CE}}(z, y) = -\sum_{j\in\{0,1\}} \omega_j \mathbb{1}[y = j] \log p_j. \tag{3.7}$$

For the training class ratio of 16:30 (real:fake) this gives $\omega_0 \approx 1.44$ and $\omega_1 \approx 0.77$. Mode **M0** (clean-only training) minimizes $\mathcal{L}_{\mathrm{CE}}$ on clean batches only.

3.7 FT-RCL detector

3.7.1 Architecture and subband signatures

The **FT-RCL detector** shares the MobileNet backbone and exposes both the classification logits and an intermediate feature map $F \in \mathbb{R}^{B\times C\times H\times W}$ (Figure 3.6). **SubbandPool** partitions the frequency axis H of the feature map into $K = 8$ bands. For band k with rows $[h_k, h_{k+1})$,

$$b_k = \mathrm{mean}_{h,w}\Big(F_{:,:,h_k:h_{k+1},:}\Big) \in \mathbb{R}^{B\times C}. \tag{3.8}$$

Stacking gives $s^{(0)} = [b_1, \ldots, b_K] \in \mathbb{R}^{B\times K\times C}$, which is L2-normalized along the channel axis:

$$s = \mathrm{normalize}\Big(s^{(0)},\ p = 2,\ \mathrm{dim} = -1\Big). \tag{3.9}$$

This gives a lightweight frequency-time summary of the internal representation, in contrast to full self-supervised embeddings.

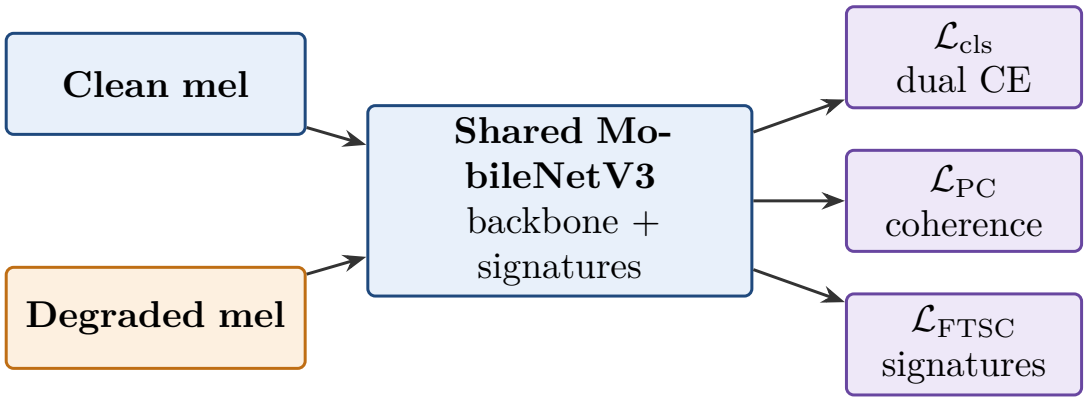


Figure 3.6. FT-RCL paired clean/degraded-view training architecture.

3.7.2 Loss functions

Let z_c, z_d be the logits from the clean and degraded mel inputs, q_c, q_d the corresponding fake-class probabilities, and s_c, s_d the signatures.

Dual classification

With the class-weighted cross-entropy of Equation (3.7),

$$\mathcal{L}_{\text{cls}} = \mathcal{L}_{\text{CE}}(z_c, y) + \beta\, \mathcal{L}_{\text{CE}}(z_d, y), \qquad \beta = 0.5. \tag{3.10}$$

Prediction coherence (PC)

$$\mathcal{L}_{\text{PC}} = \frac{1}{B} \sum_{i=1}^{B} \left(q_{c,i} - q_{d,i}\right)^2. \tag{3.11}$$

For example, suppose that for one clip the fake-class probabilities are $q_c = 0.80$ and $q_d = 0.55$. Then $(q_c - q_d)^2 = 0.0625$ contributes to $\mathcal{L}_{\text{PC}}$, and the term pushes the degraded prediction toward the clean one. If the degraded view hides high-frequency artifacts, PC penalizes the resulting overconfident disagreement. PC does not by itself guarantee a correct decision, since both views could agree on a wrong label.

Feature signature coherence (FTSC)

For band k and sample i, let $s_{c,i,k}, s_{d,i,k} \in \mathbb{R}^C$ be the L2-normalized band vectors and $\cos_{i,k} = s_{c,i,k}^{\top} s_{d,i,k}$. FTSC compares the clean and degraded signatures by the cosine similarity along the channel dimension for each band and averages over bands and batch:

$$\mathcal{L}_{\text{FTSC}} = \frac{1}{B} \sum_{i=1}^{B} \frac{1}{K} \sum_{k=1}^{K} (1 - \cos_{i,k}). \tag{3.12}$$

Figure 3.7 illustrates PC and FTSC.

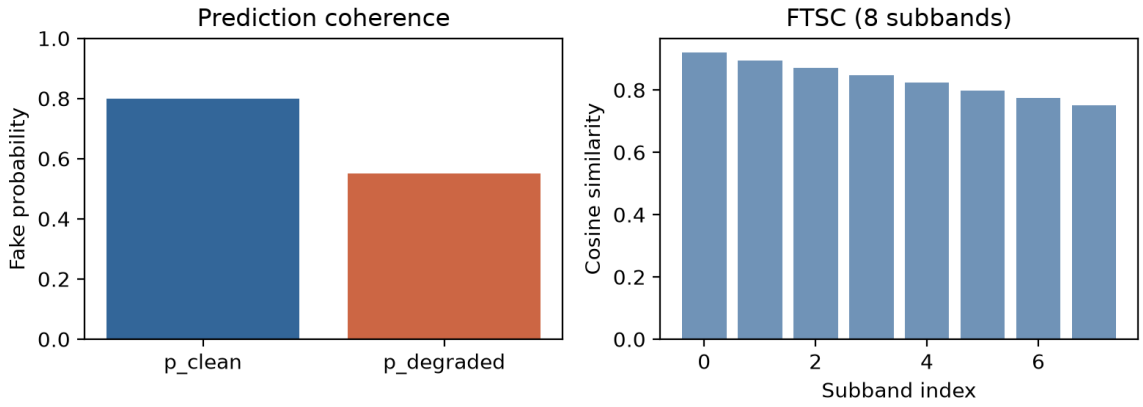


Figure 3.7. Prediction coherence and feature signature coherence.

Type, confidence, and severity robustness gate

For sample i with audio type t_i, the gate combines a type weight α_{t_i} (defaults: speech and singing 1.0, music 0.3, sound 0.5), a confidence $\kappa_i = 2|q_{c,i} - 0.5|$ (treated as a constant, so no gradient flows through κ), and the severity factor $\eta(\rho_i)$ of Equation (3.6):

$$w_i = \alpha_{t_i}\,\kappa_i\,\eta(\rho_i), \qquad \bar{w} = \frac{1}{B}\sum_{i=1}^{B} w_i. \tag{3.13}$$

The gate limits over-regularization on speech-dominated batches and lets the coherence terms act most strongly on confident, strongly degraded samples.

3.7.3 Total loss and training modes

The total loss of the full FT-RCL objective (mode M3) is

$$\mathcal{L}_{\mathrm{M3}} = \mathcal{L}_{\mathrm{cls}} + \bar{w}\left(\gamma_{\mathrm{pc}}\mathcal{L}_{\mathrm{PC}} + \gamma_{\mathrm{ftsc}}\mathcal{L}_{\mathrm{FTSC}}\right), \qquad \gamma_{\mathrm{pc}} = 1.0,\ \gamma_{\mathrm{ftsc}} = 0.5. \tag{3.14}$$

Table 3.3 summarizes how the four training modes combine the terms of Equations (3.10)–(3.12) and the gate of Equation (3.13). Mode M1 uses $\mathcal{L}_{\mathrm{CE}}(z_d, y)$ only; mode M2 uses $\mathcal{L}_{\mathrm{cls}} + \bar{w}\,\gamma_{\mathrm{pc}}\mathcal{L}_{\mathrm{PC}}$ with $\eta \equiv 1$ and $\gamma_{\mathrm{ftsc}} = 0$.

Table 3.3. Training modes and their loss components.

Mode	Network	Views used in the loss	Loss
M0	MobileNetV3-Small	Clean	$\mathcal{L}_{\mathrm{CE}}(z_c, y)$
M1	FT-RCL detector	Degraded	$\mathcal{L}_{\mathrm{CE}}(z_d, y)$
M2	FT-RCL detector	Clean and degraded	$\mathcal{L}_{\mathrm{cls}} + \bar{w}\,\gamma_{\mathrm{pc}}\mathcal{L}_{\mathrm{PC}}$ $(\eta \equiv 1)$
M3	FT-RCL detector	Clean and degraded	$\mathcal{L}_{\mathrm{cls}} + \bar{w}\,(\gamma_{\mathrm{pc}}\mathcal{L}_{\mathrm{PC}} + \gamma_{\mathrm{ftsc}}\mathcal{L}_{\mathrm{FTSC}})$

3.8 STAR-Edge extension

STAR-Edge extends the FT-RCL recipe with three additional components. It uses the same backbone, frontend, degradation catalogue, and optimization settings, and is enabled through its own configuration file.

3.8.1 Asymmetric bona-fide robustness (ABR)

ABR applies an additional robustness term to *real* clips. Genuine recordings are the class most affected when compression or noise introduces artifacts that resemble synthesis, so the term treats bona-fide clips asymmetrically to keep them from being flagged as fake after degradation.

3.8.2 Severity trajectory coherence (STC)

STC extends pairwise coherence across increasing severity levels of the same clip, so that the representation changes consistently as the degradation strengthens rather than only matching a single clean–degraded pair.

3.8.3 Channel auxiliary detection (CAD)

CAD derives an eight-dimensional channel code from waveform statistics and trains an auxiliary head to detect the channel condition. The backbone embeddings are projected orthogonally to a learned channel direction before signature extraction, which discourages the signatures from encoding channel information instead of evidence of synthesis. As with the other training-only components, CAD is not used at inference. The training progress of the STAR-Edge run is reported in Section 4.7.

3.9 Optimization and checkpoint selection

3.9.1 Optimizer and training schedule

Training uses AdamW with learning rate 10^{-3} and weight decay 10^{-4}, batch size 32, up to 20 epochs, mixed-precision training on CUDA, and early stopping with patience 5 on a selection score. The learning rate follows ReduceLROnPlateau on development Macro-F1 (factor 0.5, patience 2).

3.9.2 Checkpoint selection

Baseline. The checkpoint with the best development Macro-F1 is kept.

FT-RCL (J-score). Let $\widehat{\Delta\mathrm{F1}}_{\mathrm{dev}}$ be an estimate of the Macro-F1 drop under quick on-the-fly degradations on a subset of development batches. The selection score is

$$J = \mathrm{MacroF1}_{\mathrm{dev,clean}} - \lambda\, \widehat{\Delta\mathrm{F1}}_{\mathrm{dev}}, \qquad \lambda = 0.5, \tag{3.15}$$

and checkpoints maximize J when robustness-aware selection is enabled.

3.10 Training versus inference

Training (M3) requires paired degradations, two mel computations, dual logits and signatures, and the full loss $\mathcal{L}_{\mathrm{M3}}$. **Inference** applies ϕ once and performs a single forward pass; signatures and degradations are omitted. This preserves the latency expected of an edge-compatible deployment (Figure 3.8).

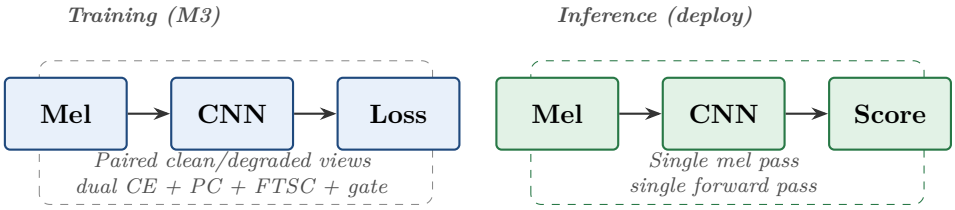


Figure 3.8. Training versus inference workflow.

3.11 Evaluation metrics

3.11.1 Metrics

For a clip with label $y \in \{0, 1\}$, the model produces logits z and probabilities $p = \mathrm{softmax}(z)$ (Section 3.2). The **detection score** is

$$q = p_1 = \frac{e^{z_1}}{e^{z_0} + e^{z_1}} \in [0, 1], \tag{3.16}$$

the predicted probability that the clip is fake. For a decision threshold τ, the predicted label is $\hat{y}_\tau = \mathbb{1}[q \geq \tau]$. Unless stated otherwise, $\tau = 0.5$. Let TP, FP, TN, and FN denote the counts of true positives, false positives, true negatives, and false negatives, with the fake class ($y = 1$) as the positive class. The metrics are defined as follows.

Accuracy: the fraction of correct decisions at threshold 0.5 on q,

$$\mathrm{Acc} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}}. \tag{3.17}$$

Macro-F1: the unweighted mean of the F1 scores of the real and fake classes. For class $c \in \{0, 1\}$, with precision P_c and recall R_c,

$$\mathrm{F1}_c = \frac{2\,P_c R_c}{P_c + R_c}, \qquad \text{Macro-F1} = \tfrac{1}{2}\left(\mathrm{F1}_0 + \mathrm{F1}_1\right), \tag{3.18}$$

where $P_1 = \mathrm{TP}/(\mathrm{TP} + \mathrm{FP})$, $R_1 = \mathrm{TP}/(\mathrm{TP} + \mathrm{FN})$, $P_0 = \mathrm{TN}/(\mathrm{TN} + \mathrm{FN})$, and $R_0 = \mathrm{TN}/(\mathrm{TN} + \mathrm{FP})$. Unless stated otherwise, the overall values are computed over all evaluation clips pooled together.

AUROC: the area under the ROC curve traced by the score q. With the false-positive rate $\mathrm{FPR}(\tau) = \mathrm{FP}/(\mathrm{FP}+\mathrm{TN})$ and the true-positive rate $\mathrm{TPR}(\tau) = \mathrm{TP}/(\mathrm{TP}+\mathrm{FN})$ at threshold τ,

$$\mathrm{AUROC} = \int_0^1 \mathrm{TPR}\Big(\mathrm{FPR}^{-1}(u)\Big)\,du = \Pr\left(q_+ > q_-\right) + \tfrac{1}{2}\Pr\left(q_+ = q_-\right), \tag{3.19}$$

where q_+ and q_- are the scores of a randomly drawn fake and real clip, respectively.

EER: the error rate at the operating point where the false-positive and false-negative rates are equal. With $\mathrm{FNR}(\tau) = \mathrm{FN}/(\mathrm{FN} + \mathrm{TP})$, it is computed at the ROC point where FPR and FNR are closest,

$$\tau^* = \arg\min_{\tau} \Big|\mathrm{FPR}(\tau) - \mathrm{FNR}(\tau)\Big|, \qquad \mathrm{EER} = \tfrac{1}{2}\Big(\mathrm{FPR}(\tau^*) + \mathrm{FNR}(\tau^*)\Big). \tag{3.20}$$

AT-ADD Macro-F1: for each audio type $t \in \mathcal{T} = \{\text{speech}, \text{singing}, \text{music}, \text{sound}\}$, the F1 scores of the real and fake classes are averaged, and the four type-wise scores are then averaged without weighting [3],

$$\mathrm{F1}_{\text{AT-ADD}} = \frac{1}{|\mathcal{T}|}\sum_{t\in\mathcal{T}} \tfrac{1}{2}\left(\mathrm{F1}_0^{(t)} + \mathrm{F1}_1^{(t)}\right), \tag{3.21}$$

where $\mathrm{F1}_c^{(t)}$ is the F1 score of class c computed on the clips of type t only. Reported results use clean held-out evaluation clips only; no hyperparameters are tuned on evaluation data.

Chapter 4

Results and Evaluation

4.1 Foundational Baseline Investigation and Architectural Selection

Before evaluating the proposed FT-RCL robustness framework, we conducted an empirical investigation into foundational baseline architectures and acoustic frontend configurations. This study addressed three core design questions in systematic order:

1. How do frontend acoustic sampling rates and duration normalization policies affect detection validity and vulnerability to benchmark shortcuts?
2. Which lightweight backbone paradigm is best suited for heterogeneous all-type audio: pure 2D convolutional networks or hybrid convolutional-recurrent architectures?
3. How does our tailored lightweight baseline compare against published literature on the same backbone architecture?

4.1.1 Acoustic Frontend Engineering and ASVspoof 2019 Duration Shortcut Analysis

A critical finding from our preliminary speech benchmarking on ASVspoof 2019 Logical Access [2] concerned the choice of duration normalization and acoustic sampling bandwidth:

- **Duration Shortcuts under Zero-Padding:** When utterances are zero-padded to a fixed 4.0-second window, classifiers readily exploit differences in active speech duration and trailing silence ratios as an unintended benchmark shortcut. On ASVspoof 2019 LA, evaluating under a 16 kHz zero-padded frontend yielded an artificially low evaluation EER of 6.67% (E_1 v1) and an attack detection recall exceeding 98.6% on the i-vector/PLDA voice conversion system (A18). However, this high performance collapsed when evaluating continuous speech where trailing silence is absent.
- **Circular Repeat-Padding as a Fair Evaluator:** Replacing trailing silence with circular repeat-padding ($\mathbf{x}_{\text{padded}}[n] = \mathbf{x}[n \pmod L]$) maintains continuous acoustic energy across the entire 4.0-second window. On the same benchmark, evaluation EER under repeat-padding adjusted to 12.49% (E_1 v2), and recall on

A18 dropped to 0.43%. While the numerical error increased, this performance reflects true acoustic discrimination of vocoder artifacts rather than duration shortcuts.

- **Matched Speech Performance (Pure CNN vs. CRNN):** Crucially, when evaluating under strictly matched preprocessing pipelines on ASVspoof 2019 LA speech, pure 2D CNN (E_1) consistently matches or slightly outperforms CRNN (E_2): 6.67% vs. 6.82% EER at 16 kHz zero-pad, and 12.49% vs. 12.59% EER at 24 kHz repeat-pad (with E_2-GRU reaching 13.82%). This demonstrates that temporal recurrence provides no intrinsic advantage over spatial pooling on speech when frontend confounding is removed.
- **Sampling Bandwidth (16 kHz vs. 24 kHz):** Operating at a 24 kHz sampling rate (12 kHz Nyquist) is critical for capturing high-frequency artifacts produced by modern neural vocoders (e.g., transposed-convolution checkerboard patterns in HiFi-GAN and BigVGAN in the 8–12 kHz band), which are truncated by standard 16 kHz telephonic frontends.

Consequently, the 24 kHz circular repeat-padding frontend was adopted as the uniform standard across all AT-ADD Track 2 models, ensuring that robustness gains are not confounded by duration artifacts.

4.1.2 Backbone Architecture Ablation: Pure 2D CNN vs. CRNN on Multi-Domain Audio

Having established the 24 kHz circular repeat-padding frontend, we evaluated two architectural paradigms under identical training protocols on the complete AT-ADD Track 2 benchmark to identify the optimal backbone for edge-constrained detection (≤ 1.52 M parameters):

- E_1 **Series (Pure 2D CNN):** MobileNetV3-Small [18] with depthwise separable convolutions, Squeeze-and-Excitation (SE) channel attention, and spatial Global Average Pooling (GAP) collapsing the spatial feature map into a 576-dimensional embedding before a two-stage classification head (1,519,618 total parameters).
- E_2 **Series (CRNN):** The same MobileNetV3-Small convolutional feature extractor, but retaining the temporal sequence axis via frequency-only pooling ($\mathbb{R}^{B\times 576\times T'\times F'} \rightarrow \mathbb{R}^{B\times T'\times 576}$) and passing the sequence through recurrent temporal pooling heads: a 1-layer GRU [36] (1,198,082 parameters) and a 1-layer LSTM [37] (1,288,450 parameters) with hidden dimension 128.

Table 4.1 summarizes the official evaluation metrics on the full AT-ADD Track 2 evaluation partition.

The empirical results show that the pure 2D CNN with spatial global average pool-

Table 4.1. Baseline architecture comparison on full AT-ADD Track 2 evaluation partition.

Model Identifier	Architecture	Params	Accuracy (%)	AUROC	EER (%)
AT-ADD E_1 v4	Pure 2D CNN (Spatial GAP)	1.52 M	**75.70%**	**0.8750**	**19.92%**
AT-ADD E_2 GRU v4	CRNN (1-Layer GRU, 128-dim)	1.20 M	69.89%	0.8425	22.43%
AT-ADD E_2 LSTM v4	CRNN (1-Layer LSTM, 128-dim)	1.29 M	69.60%	0.8622	22.15%

ing (E_1 v4) significantly outperforms both recurrent architectures on multi-domain audio, reducing the Equal Error Rate by over 2.2% absolute (19.92% vs. 22.15% for LSTM and 22.43% for GRU) and achieving nearly 6% higher classification accuracy. In heterogeneous audio containing polyphonic music, percussion, and diverse ambient soundscapes, local temporal ordering lacks the strict phoneme-to-phoneme transition dynamics of speech. Spatial average pooling functions as a robust "bag-of-features" aggregator across time-frequency bins, whereas recurrent transitions overfit to non-stationary acoustic sequences. This empirical ablation directly established MobileNetV3-Small (E_1) as the optimal lightweight backbone for all subsequent experiments, including Baseline M0 and FT-RCL M3.

Furthermore, Table 4.1 establishes the full-data performance ceiling (19.92% EER, 75.70% accuracy) against which the accelerated 30% split protocol (21.6% EER, 75.9% accuracy in Section 4.4) can be calibrated.

4.1.3 Empirical Literature Benchmarking against Published Work (Pham et al., 2024)

To contextualize our lightweight baseline against existing literature, we conducted an independent reproduction and benchmarking study against Pham et al. (2024) [19], who evaluated ImageNet-pretrained 2D vision backbones on ASVspoof 2019 LA. In their study, System C2 fine-tuned a MobileNet-V3 backbone on 3-channel linear filterbank spectrograms ($64 \times 64 \times 3$, static LFCC + Δ + $\Delta\Delta$ extracted from 2.0-second slices with frame length $N_{\text{FFT}} = 1024$ and hop 512, corresponding to a 64 ms window and 32 ms shift at 16 kHz).

We faithfully reconstructed their exact signal processing chain, network configuration, and segment-to-utterance probability pooling under our standardized optimization recipe (AdamW, cosine learning rate decay, class-weighted cross-entropy, and per-bin normalization). Table 4.2 compares published results, our reproduced run, and our project baselines on the official ASVspoof 2019 LA evaluation partition.

This comparison yields two critical insights:

1. **Optimization Sensitivity:** In the original paper by Pham et al., System C2 reached 48.00% EER, performing barely above a random coin flip. Our independent reproduction stabilized the training dynamics, bringing the exact same

Table 4.2. Empirical comparison: published literature (Pham et al., 2024) vs. reproduced run and project baselines on ASVspoof 2019 LA evaluation partition.

System / Model	Acoustic Frontend	Accuracy (%)	F1 / Macro-F1	AUROC	EER (%)
Tier A: Published in Pham et al. (2024) Table III:					
Pham et al. C2 (MobileNet-V3)	STFT & LF ($64 \times 64 \times 3$)	59.00%	0.6700*	0.5200	48.00%
Pham et al. C1 (ResNet-18)	STFT & LF ($64 \times 64 \times 3$)	49.00%	0.5800*	0.5100	47.00%
Pham et al. C7 (ConvNeXt-Tiny, 28.6 M params)	STFT & LF ($64 \times 64 \times 3$)	88.00%	0.9000*	0.9600	7.50%
Pham et al. Best Ensemble (Multi-feature)	STFT, CQT, LF, Gammatone	90.00%	0.9100*	0.9940	**3.00%**
Tier B: Our Independent Reproduction Run:					
Replicated Pham C2 (Segment-level)	STFT & LF ($64 \times 64 \times 3$)	64.95%	0.5636	0.9162	17.01%
Replicated Pham C2 (Utterance-level)	STFT & LF ($64 \times 64 \times 3$)	**64.60%**	**0.5612**	**0.9180**	**16.85%**
Tier C: Our Tailored Audio Baselines (This Project):					
Project E_1 v2 (Pure CNN)	80-Mel, 24 kHz, Repeat-Pad	**75.32%**	**0.6478**	**0.9324**	**12.49%**
Project E_1 v1 (Pure CNN)	80-Mel, 16 kHz, Zero-Pad	**90.90%**	**0.8196**	**0.9688**	**6.67%**

**Pham et al. report majority-class / weighted F1, whereas Tier B and C report unweighted Macro-F1.*

MobileNet-V3 backbone to **16.85% EER** and 0.9180 AUROC. This demonstrates that MobileNet-V3 is capable of learning discriminative representations from time-frequency audio when optimization is properly controlled.

2. **Acoustic Frontend Superiority:** Our tailored full-utterance log-Mel frontend (E_1 v2: 12.49% EER under repeat-padding) substantially outperforms the 2.0-second sliced LFCC representation (16.85% EER) while operating under the same ≤ 1.52 M parameter budget, confirming the efficacy of our baseline pipeline.

4.2 Robustness Coherence Learning Protocol (AT-ADD Track 2 Evaluation)

Having established MobileNetV3-Small (E_1) with the 24 kHz repeat-padded frontend as our canonical lightweight baseline, we evaluate the proposed FT-RCL framework against this baseline under the AT-ADD Track 2 robustness benchmark. All subsequent experiments in this chapter share the following standardized protocol:

- **Dataset:** AT-ADD Track 2 (seed 42); see Table 3.1.
- **Preprocessing:** 24 kHz, four-second segments, repeat-pad policy, 80-bin log-mel, global normalization with the E1 v4 training statistics.
- **Baseline model:** MobileNetV3-Small trained on clean audio only (mode M0).
- **Proposed model:** FT-RCL with the full M3 objective of Equation (3.14) (paired degradations, PC, FTSC, gating, J-score selection).
- **Evaluation:** held-out evaluation clips only; no hyperparameter tuning on evaluation data.

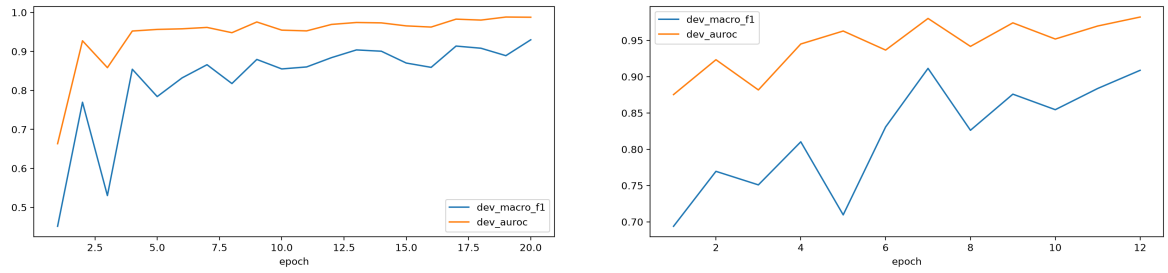
4.3 Training behaviour

Table 4.3 summarizes the peak development metrics. The baseline selects checkpoints by development Macro-F1, whereas FT-RCL uses the J-score of Equation (3.15), which penalizes the estimated robustness drop on degraded development batches. The baseline reaches its peak at epoch 20, the last epoch of the training budget, so its development score may not yet have saturated and longer training could change the baseline numbers.

Table 4.3. Peak development metrics during training.

Model	Dev Macro-F1	Dev EER	Selection score
Baseline (M0)	0.930	0.053	0.930 (Macro-F1)
FT-RCL (M3)	0.909	0.068	0.902 (best J)

FT-RCL reached high training accuracy (>96%) and strong development Macro-F1 (Figure 4.1). The logged training loss of M3 was numerically unstable under mixed precision (Figure 4.2), so model selection relied on development Macro-F1 and J instead of the training loss. Figure 4.3 shows the development EER of both models.



(a) Baseline (M0): peak dev Macro-F1 ≈0.930 at epoch 20

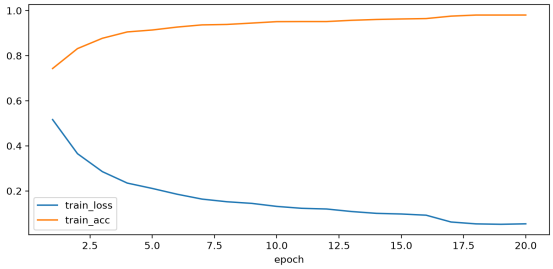
(b) FT-RCL (M3): peak dev Macro-F1 ≈0.909, best J≈0.902 at epoch 12

Figure 4.1. Development Macro-F1 and AUROC during training.

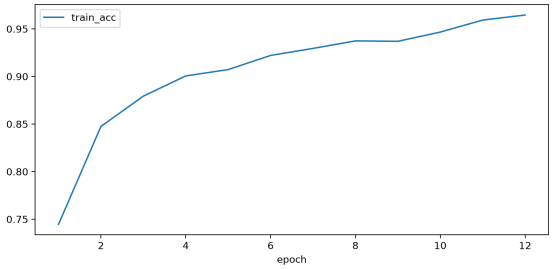
4.4 Clean evaluation (primary results)

Table 4.4 reports the metrics on **clean** evaluation audio (no synthetic degradation at test time), and Figure 4.4 compares the two systems graphically.

On the evaluation set the baseline slightly exceeds FT-RCL in aggregate Macro-F1, accuracy, AUROC, and EER. FT-RCL optimizes additional robustness terms during training, which can trade off against aggregate clean scores; clean evaluation alone does not test compression robustness, and conditioned evaluation is reported in Section 4.6.

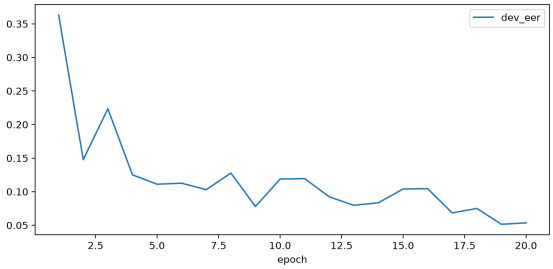


(a) Baseline: training loss and accuracy

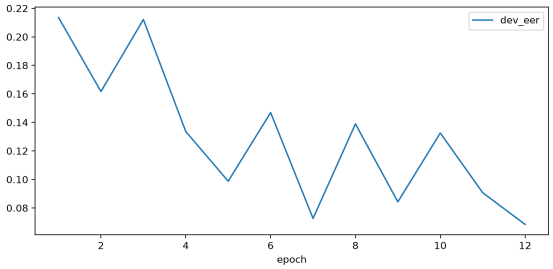


(b) FT-RCL: training accuracy (loss unreliable under mixed precision)

Figure 4.2. Training loss and accuracy during training.



(a) Baseline: dev EER versus epoch



(b) FT-RCL: dev EER versus epoch

Figure 4.3. Development EER during training.

Table 4.4. Overall clean evaluation metrics.

Model	Macro-F1	Accuracy	AUROC	EER
Baseline (E1 v4, M0)	0.672	0.759	0.852	0.216
FT-RCL (M3)	0.659	0.748	0.843	0.234

The ROC curves and the confusion matrices at threshold 0.5 are shown in Figures 4.5 and 4.6.

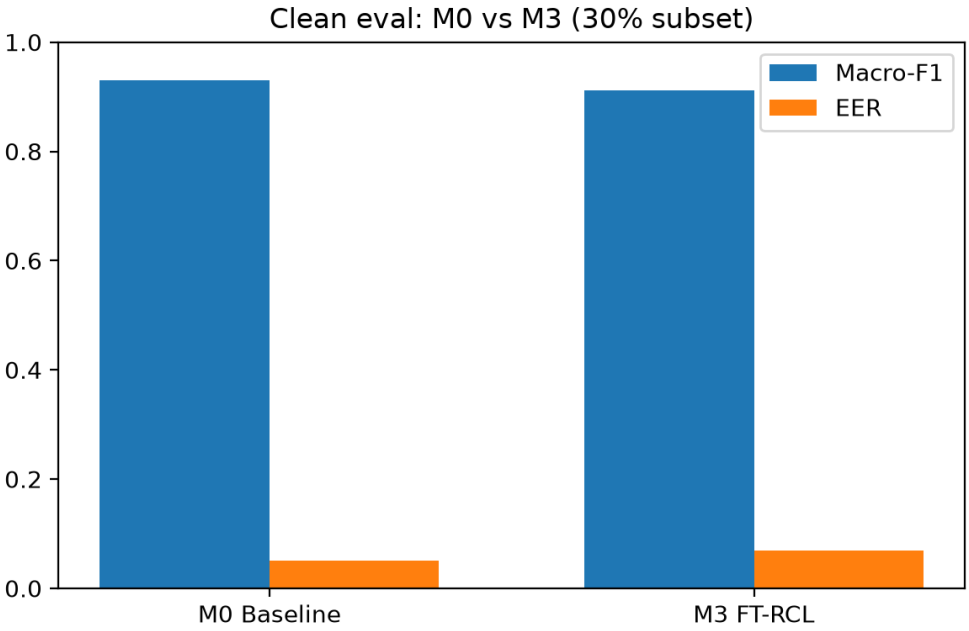
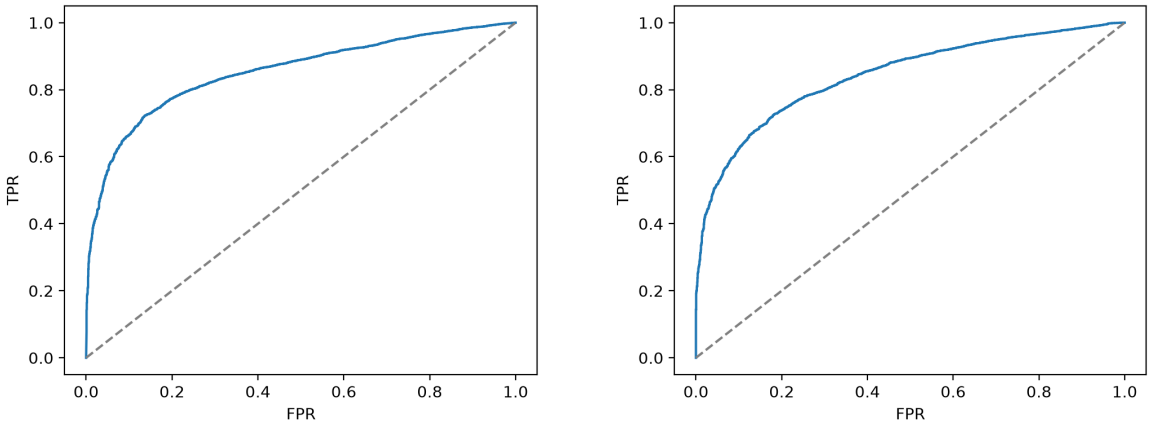


Figure 4.4. Clean-evaluation comparison of baseline M0 and FT-RCL M3.



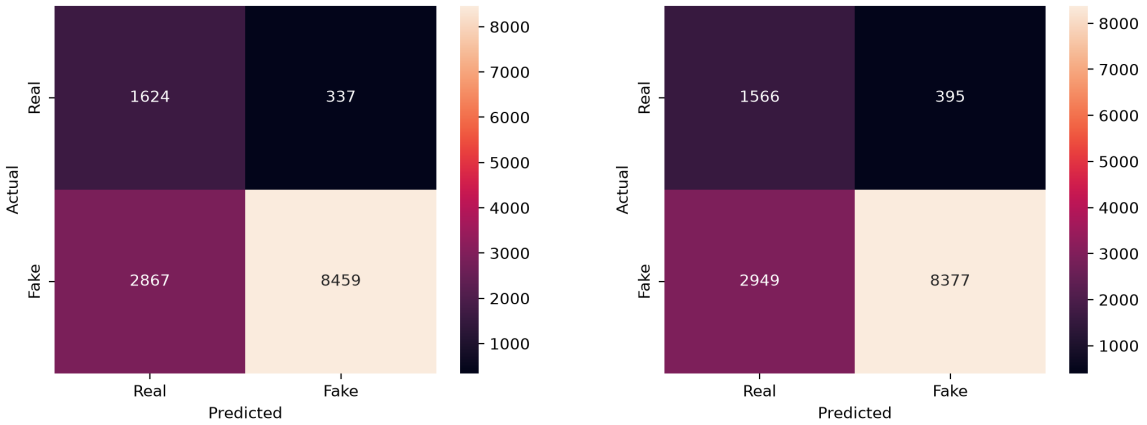
(a) Baseline: AUROC 0.852

(b) FT-RCL: AUROC 0.843

Figure 4.5. ROC curves on clean evaluation (fake-class score).

4.5 Per-audio-type analysis

Table 4.5 breaks the clean-evaluation performance down by audio type. Both models struggle on **music**. Relative to the baseline, FT-RCL improves Macro-F1 on **sound** and **music**, with trade-offs on speech and singing.



(a) Baseline (M0)

(b) FT-RCL (M3)

Figure 4.6. Clean-evaluation confusion matrices at threshold 0.5.

Table 4.5. Macro-F1 and EER by audio type (clean evaluation).

Type	Baseline (M0)		FT-RCL (M3)	
	Macro-F1	EER	Macro-F1	EER
Speech	0.695	0.162	0.665	0.179
Singing	0.794	0.157	0.764	0.208
Sound	0.467	0.426	0.522	0.394
Music	0.393	0.399	0.403	0.467
Type-averaged (AT-ADD) Macro-F1	0.587		0.589	

The last row of Table 4.5 gives the type-averaged Macro-F1 that corresponds to the AT-ADD Track 2 definition. It is the unweighted mean of the four rows above and was computed from the rounded table entries. It is much lower than the pooled value in Table 4.4 because music and sound pull the average down, and the two systems are essentially tied on it (0.587 versus 0.589). This pronounced difficulty on music directly corroborates our foundational full-split baseline ablation (E_1 v4 in Section 4.1.2), where the music partition exhibited an EER of 46.09% and Macro-F1 of 0.3907, confirming that music classification is an intrinsic domain difficulty across both full-data and 30% subsampled training regimes rather than a sampling artifact.

4.6 Degraded-audio and robustness evaluation

The conditioned evaluation reported here is **preliminary**. It was run on a small set of evaluation clips ($n = 100$ clips per model), and MP3 conditions are omitted when the codec tool is unavailable.

Table 4.6. Mean ΔMacro-F1 (clean minus conditioned) by degradation condition on an eval subsample.

Condition	Severity	Baseline M0	FT-RCL M3
bandlimit_300_3400	3	0.115	0.134
down_up_16k	1	0.000	0.000
snr_10db	3	-0.320	-0.262
snr_20db	2	-0.310	-0.223
snr_5db	4	-0.320	-0.268

Figure 4.7 shows the conditioned Macro-F1 and ΔMacro-F1 against degradation severity, and Figure 4.8 shows the ΔMacro-F1 heatmap by audio type and condition. This set contains at most 100 clips in total and therefore far fewer per audio type, so individual cells are dominated by sampling noise. These figures demonstrate the conditioned-evaluation pipeline and are not used to rank the two systems.

4.7 STAR-Edge extension

STAR-Edge extends FT-RCL with the ABR, STC, and CAD components described in Section 3.8. The extension was implemented and trained under the same data, preprocessing, and optimization settings as the other systems, and its development dynamics are shown in Figure 4.9. A held-out clean evaluation of the STAR-Edge checkpoint under the protocol of Table 4.4 is not part of the comparison above; it is listed in the future work (Chapter 5).

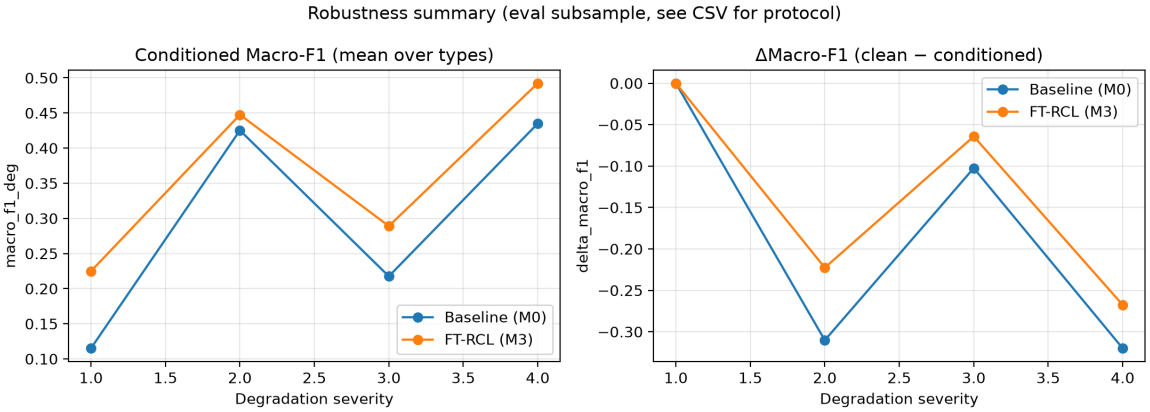
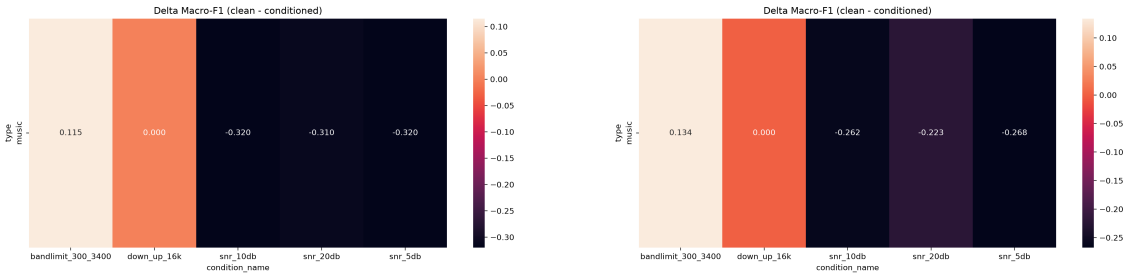
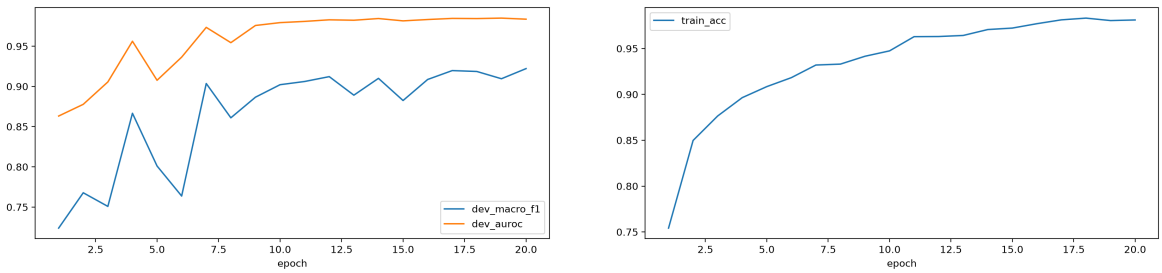


Figure 4.7. Conditioned Macro-F1 and ΔMacro-F1 versus degradation severity (n=100 clips per model; MP3 omitted if the codec tool is unavailable).



(a) Baseline (M0) (b) FT-RCL (M3)

Figure 4.8. ΔMacro-F1 heatmap by audio type and condition (n=100 clips per model).



(a) Development Macro-F1 and AUROC (b) Training accuracy

Figure 4.9. STAR-Edge training progress.

4.8 Edge and efficiency

Table 4.7 summarizes stored checkpoint size for the baseline and FT-RCL (∼17.6 MB each, ∼1.52 M parameters). This is only a proxy for the deployment footprint; inference latency, peak memory, and INT8 accuracy have not been measured.

Table 4.7. Model size summary (inference latency and INT8 accuracy pending edge benchmark).

Model	Parameters	Checkpoint (MB)
Baseline (M0)	1,519,618	17.60
FT-RCL (M3)	1,519,618	17.61

4.9 Analysis and Interpretation

Aggregate clean metrics: On the evaluation set the baseline exceeds FT-RCL in Macro-F1 (0.672 versus 0.659), accuracy, AUROC, and EER (Table 4.4). FT-RCL optimizes auxiliary coherence terms, and its J-score explicitly penalizes the estimated development-time robustness drop, which can move the selected checkpoint away from the clean-optimal point. Without complete conditioned test metrics, it is not possible to say whether this trade-off improves ΔM.

Per-type patterns: Table 4.5 shows FT-RCL improving Macro-F1 on environmental sound (+0.055) and music (+0.010) while lowering it on speech and singing (−0.030 each). The type-averaged Macro-F1 is essentially identical (0.587 versus 0.589), so the gains on the harder domains offset the losses on the easier ones. EER does not move in lockstep with Macro-F1: the music EER is worse for FT-RCL (0.467 versus 0.399) even though its F1 is slightly higher. Macro-F1 depends on a fixed threshold, whereas EER summarizes score ranking across thresholds, so such discrepancies are expected when score distributions shift without a uniform benefit.

Development versus evaluation gap: Both models reach high development Macro-F1 (≈0.93 for the baseline and ≈0.91 for FT-RCL at their best epochs) but a lower evaluation Macro-F1 (≈0.66–0.67). The gap reflects possible overfitting to the development data and the difficulty of the generator-disjoint evaluation partition. Claims should therefore rest on the held-out evaluation tables and not on the training curves alone (Figures 4.1–4.3).

Statistical caution: A single training run with one seed does not support formal significance testing. The observed differences may change under longer training, more data, or repeated seeds, and the report therefore states **observed differences**, not proven superiority.

Robustness and edge claims. The conditioned results (Table 4.6 and Figures 4.7 and 4.8) come from a small number of clips, and conditioned evaluation on the complete evaluation set is required before any robustness claim. Edge suitability additionally requires latency and INT8 measurements beyond Table 4.7.

Edge deployment: Stored checkpoint size is similar for both models (Table 4.7); latency, peak memory, and INT8 accuracy on a target device remain to be measured.

4.10 Principal Outcomes

1. Both systems achieve usable speech and singing detection on the evaluation set (EER 0.16–0.21); **music** remains the primary failure mode (EER $\approx$ 0.40–0.47), and sound is also difficult (EER 0.39–0.43).
2. FT-RCL M3 does not surpass the baseline on **aggregate clean** metrics in these experiments. Its gains on sound and music are partial, are accompanied by mixed EER behaviour, and leave the type-averaged Macro-F1 unchanged.
3. The high development scores of both models, contrasted with their lower evaluation scores, call for larger-scale training and degraded-audio evaluation before conclusions about robustness are drawn.
4. Conditioned ΔM evaluation on the complete evaluation set and the M1/M2 ablations are required to test the core FT-RCL motivation; the preliminary results of Figures 4.7 and 4.8 do not settle it.
5. Held-out STAR-Edge clean evaluation remains a useful next step alongside the ablations above.

Chapter 5

Conclusion and Future Work

5.1 Conclusion

5.1.1 Summary

This project implemented lightweight binary audio deepfake detection for AT-ADD Track 2 using log-mel features and MobileNetV3-Small, together with FT-RCL robustness training: paired clean/degraded views, prediction and signature coherence, a severity curriculum, and gated robustness losses. The STAR-Edge extension, with asymmetric bona-fide robustness, severity trajectory coherence, and channel auxiliary detection, was also implemented and trained. In every case the inference graph is that of the baseline: one mel computation and one forward pass of a $\sim$1.52 M-parameter network.

On clean evaluation audio, the baseline achieved Macro-F1 0.672 and EER 0.216, and FT-RCL reached Macro-F1 0.659 and EER 0.234 under the same protocol. Both systems give usable detection on speech and singing (EER 0.16–0.21) but struggle on music and environmental sound (EER 0.39–0.47). FT-RCL gave slightly better Macro-F1 on environmental sound and music but no uniform gain on aggregate clean metrics, and the type-averaged Macro-F1 of the two systems is essentially the same (0.587 versus 0.589). The results confirm that all-type detection remains difficult for compact CNNs, especially on music, in line with the AT-ADD benchmark findings [3].

Primary conclusions rest on clean held-out evaluation (13,287 clips). Section 4.6 adds a preliminary conditioned ΔM study on 100 clips per model, which validates the evaluation pipeline but is too small for robustness ranking. The M1/M2 ablations and the held-out STAR-Edge evaluation have not been run, and on-device latency and quantization have not been measured. The report therefore claims observed differences on clean audio and does not claim proven compression robustness or edge suitability.

5.1.2 Limitations

- **Training scale and statistical power.** Each system was trained once with a single seed under a student-scale compute budget, so the observed differences cannot be tested for significance and may change with longer training or additional data.

- **Limited conditioned evaluation:** Section 4.6 uses n=100 clips per model; full eval-set ΔM and MP3 conditions (when ffmpeg is missing) are incomplete [5], [22].
- **Speaker and source leakage** between partitions cannot be fully verified from the available AT-ADD metadata.
- **MP3 degradations** depend on an external codec toolchain; if it is unavailable, the severity ladder is incomplete.
- **Edge deployment** (quantization, latency, memory on a device) is not validated in this phase [6], [25].
- **STAR-Edge** has been trained, but its held-out evaluation is not included in the primary result tables.

5.2 Future work

1. **Larger-scale training and ΔM.** Train and evaluate the baseline and FT-RCL on the complete evaluation split with the full degradation catalog; repeat with several seeds for significance testing.
2. **Ablations and STAR-Edge evaluation.** Complete the M1/M2 ablations to separate the contributions of prediction coherence, signature coherence, and the robustness gate; evaluate the STAR-Edge checkpoint under the same clean held-out protocol; and compare J-score checkpointing with development-Macro-F1 checkpointing.
3. **Training stability.** Stabilize the training-loss logging (higher-precision loss computation) for reliable FT-RCL monitoring under mixed precision.
4. **Edge benchmarking.** Execute an edge benchmark: parameter count, checkpoint size, batch-1 latency on a defined device, peak memory, and INT8 quantization accuracy.
5. **Failure analysis.** Analyse failure cases by generator, especially music text-to-music systems, to guide targeted robustness or architecture changes.

Appendix A

Implementation Details

A.1 Configuration reference

Key fields of the FT-RCL configuration, stored in `configs/proposed/e7_ft_rcl_atadd_v1.yaml`:

- **Preprocessing:** version `v4.0_24khz_repeat_pad`; mel statistics stored in `E1_ATADD_logmel_stats_v4.json`.
- **Training:** batch size 32, 20 epochs, AdamW with learning rate 10^{-3} and weight decay 10^{-4}, early-stopping patience 5, seed 42.
- **Selection:** `use_j_score: true`, $\lambda = 0.5$, `delta_max_batches: 5`.
- **FT-RCL:** $\beta = 0.5$, $\gamma_{\text{pc}} = 1.0$, $\gamma_{\text{ftsc}} = 0.5$; type weights α_t: speech and singing 1.0, music 0.3, sound 0.5; curriculum epochs $[1, 6, 13] \rightarrow$ maximum severities $[1, 3, 4]$.
- **STAR-Edge:** all flags `false` in this configuration; the STAR-Edge run uses a separate configuration file.

A.2 Training-loop pseudocode

The following pseudocode outlines one training run for modes M1–M3 (Section 3.7.2).

```
for epoch in 1..E:
  set_max_severity(curriculum(epoch))
  for clean_w, deg_w, y, type, meta in train_loader:
    if mode == M1:
      loss = CE(model(mel(deg_w)), y)
    elif mode in {M2, M3}:
      zc, sc, _ = model(mel(clean_w), waveforms=clean_w)
      zd, sd, _ = model(mel(deg_w),   waveforms=deg_w)
      eta  = mean(severity/meta) if M3 else 1.0
      loss = ft_rcl_loss(zc, zd, y, sc, sd, type, eta, ...)
    backward(loss); AdamW.step()
  save checkpoint if J-score improves
```

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